



King Abdul Aziz University

Faculty of Computing & Information Technology
Computer Science Department

Exam 1 (July 25th, 2017)

Course: CPCS-223 (Analysis and Design of Algorithm)

Time: 60 minutes

Student ID: _____

Student Name: _____

***Section: _____

Instructor:

☐ Dr Umair

☐

Question 01

Which of the followings is / are not the application of the Graph?

- a) Travelling Sales Problem b) Computer Network
c) Air Flight Network d) None

Question 02

How many linked lists are used to represent a graph with n nodes and m edges, when using an adjacency-list representation?

- a) m b) n
c) $n*m$ d) $n+m$

Question 03

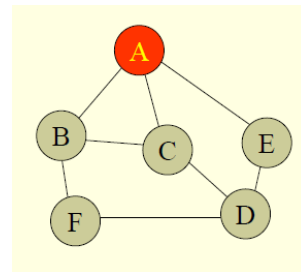
What will be the order of matrix to represent a graph with m nodes and n edges, when using an adjacency-matrix representation?

- a) $n * n$ b) $m * n$
c) $n*m$ d) $m * m$

Question 04

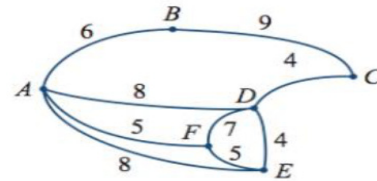
What will be the “in-degree” of vertex “C” in given graph?

- a) 2 b) 3
c) 4 d) None

**Question 05**

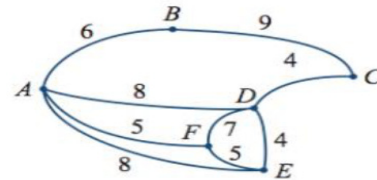
Perform a **Depth-First Traversal** of the weighted graph shown below; starting with vertex A. Select the **largest-weight** edge first, when accessing neighbors of a vertex. What is the order in which the vertices are visited?

- a) AFDECB b) AFEDCB
c) ABCDEF d) None

**Question 06**

Perform a **Breadth-First Traversal** of the weighted graph shown below; starting with vertex A. Select the **Smallest-weight** edge first, when accessing neighbors of a vertex. What is the order in which the vertices are visited?

- a) AFDECB b) AFEDCB
c) ABCDEF d) None

**Question 07**

A complete graph with 5 vertices has how many edges

- a) 5 b) 8
c) 10 d) None

Question 08

Multiple occurrence of the “same edge” is called:

- a) Self Edge b) Multi Graph
c) Cycle d) None

Question 09

Maximum number of edges in a “**Directed Graph**” is calculated as:

- a) $n(n-1)/2$
- b) $n(n-1)$
- c) $n(n+1)/2$
- d) $n(n+1)$

Question 10

Maximum number of edges in a “**Un-Directed Graph**” is calculated as:

- a) $n(n-1)/2$
- b) $n(n-1)$
- c) $n(n+1)/2$
- d) $n(n+1)$

Question 11

Every “connected vertices” are also “adjacent vertices” always.

- a) True
- b) False

Question 12

In “Simple Path” all vertices will be distinct **except**:

- a) First
- b) Last
- c) First & Last both
- d) None

Question 13

How much time it will take to find out whether vertex U is adjacent to vertex V, if “**Adjacency Matrix**” is used?

- a) n
- b) 1
- c) n^2
- d) None

Question 14

How much time it will take to find out whether vertex U is adjacent to vertex V, if “**Adjacency List**” is used?

- a) n
- b) 1
- c) n^2
- d) None

Question 15

Assuming that “root” of max heap is at index “1”, how the value of parent (i) may be calculated?

- a) $(i-1)/2$
- b) $i-1/2$
- c) $i/2$
- d) None of above

Question 16

The condition to check **Max-Heap** property is:

- a) $A[\text{Parent}(i)] \leq A[i]$
- b) $A[\text{Parent}(i)] \neq A[i]$
- c) $A[\text{left}(i)] \leq A[i]$
- d) $A[\text{Parent}(i)] \geq A[i]$

Question 17

The condition to check **Min-Heap** property is:

- a) $A[\text{Parent}(i)] \leq A[i]$
- b) $A[\text{Parent}(i)] \neq A[i]$
- c) $A[\text{left}(i)] \leq A[i]$
- d) $A[\text{Parent}(i)] \geq A[i]$

Question 18

What is the running time of “Max-Heapify”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 19

What is the running time of “Build-Max-Heap”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 20

What is the running time of “Heap-Sort”?

- a) n
- b) $\lg n$
- c) $n \lg n$
- d) n^2

Question 21

Which of the following is not a characteristic of “Hashing”?

- a) Searching Technique
- b) No Comparison of “key” with values
- c) Running time is constant
- d) None

Question 22

Which of the following techniques use “**hash function**” to find the “**hash address**”?

- a) Direct Address Table
- b) Hash Table
- c) Open Addressing
- d) Both b & c

Question 23

Probing is a technique to address the collision problem for:

- a) Direct Address Table
- b) Hash Table
- c) Open Addressing
- d) None

Question 24

Chaining is a technique to address the collision problem for:

- a) Direct Address Table
- b) Hash Table
- c) Open Addressing
- d) None

Question 25

Problem Statement specifies the relationship between:

- a) Input & Process
- b) Process & Output
- c) Output & Input
- d) None

Question 26

Which of the following is / are not the basic characteristic of an **Algorithm**?

- a) Step by Step Method
- b) Unambiguous Instruction
- c) Computational Procedure
- d) Instances

Question 27

Which of the followings are the designing technique(s) of algorithm?

- a) Greedy Algorithm
- b) Decrease and Conquer
- c) Transform and Conquer
- d) All of above

Question 28

There are TWO phases in “problem solving”, which are:

- a) Analysis & Design
- b) Analysis & Synthesis
- c) Design & Synthesis
- d) None

Question 29

Which of the following is not the part of “Synthesis” phase in solving a problem?

- a) Building functions with supporting tools
- b) How components should be put together?
- c) How components (processes) relate to each other
- d) Composing functions to form a process

Question 30

How “Time” and “Space” are related to each other?

- a) Directly Proportional
- b) Inversely Proportional
- c) Both ways
- d) None

Question 31

Efficiency claim about an algorithm must be independent of

- a) Number of Instances
- b) Bench Marks
- c) Both a & b
- d) None

Question 32

While describing the efficiency of an algorithm, which term(s) are not ignored at all?

- a) Highest Order term
- b) Lower Order term
- c) All Constant
- d) All of Above

Question 33

Which of the following “Complexity Class” represents the **minimum** running time?

- a) $n \lg n$
- b) $\lg n$
- c) n
- d) n^2

Question 34

The running time of “Radix Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 35

The running time of “Selection Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 36

The running time of “Merge Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 37

The running time of “Quick Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 38

What is the percentage weightage of your “Final exam” of CPCS-223?

- a) 20
- b) 30
- c) 35
- d) 40

Question 39

On which date “exam 2” of CPCS-223 will be conducted according to given schedule?

- a) Tuesday, July 25, 2017
- b) Tuesday, August 08, 2017
- c) Tuesday, August 15, 2017
- d) There will be no “Exam 2”

Question 40

For RAM model “RAM” stands for:

- a) Random Analysis Machine
- b) Random Access Memory
- c) Random Analysis Memory
- d) Random Access Machine

Question 41

RAM model is designed primarily for

- a) Single Processor Machine
- b) Multi Processor Machine
- c) Both a & b
- d) None

Question 42

For RAM model, which one is not the basic assumption of the model?

- a) Infinitely large random-access memory
- b) Every instruction completes its execution in “unit” time
- c) Instructions do not execute sequentially
- d) Reproducible execution time

Question 43

Which analyses produce the same results most of the time?

- a) Best Case & Average Case
- b) Average Case & Worst Case
- c) Worst Case & Best Case
- d) None

Question 44

What will be the value of $\sum_{k=0}^n m$

- a) $n * m$
- b) n
- c) $n + m$
- d) $(n + 1) * m$

Question 45

What will be the value of $\sum_{k=4}^n 1$

- a) $n - 3$
- b) n
- c) $n - 4$
- d) None

Question 46

In “Worst Case” analysis, the running time of “Insertion Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 47

In “Best Case” analysis, the running time of “Insertion Sort” is:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(n \lg n)$
- d) $O(\lg n)$

Question 48

Asymptotic notation describes the running time variations between “Upper” bound and “Lower” bound:

- a) True
- b) False

Question 49

Which of these is / are not “tight bounds”?

- a) o
- b) Θ
- c) Ω
- d) ω

Question 50

According to the definition, which condition(s) is / are correct for Theta-Notation?

- a) $f(n) > 0$
- b) c_1 and c_2 must be positive integers
- c) n can be negative
- d) It describes Upper Bound

Question 51

According to the definition, which condition(s) is / are not correct for Omega-Notation?

- a) $f(n) > 0$
- b) It is tight bound
- c) c must be a positive constant
- d) It describes Upper Bound only

Question 52

If $\frac{1}{2}n^2 - \frac{1}{2}n = \Theta(n^2)$ for $c_1 = \frac{1}{4}$ and $c_2 = \frac{1}{2}$, what will be the value of n_0 ?

- a) $n_0 > 2$
- b) $n_0 \geq 2$
- c) $n_0 < 2$
- d) $n_0 \leq 2$

Question 53

“Loop Invariant” is used for defining

- a) Algorithm’s Efficiency
- b) Algorithm’s Correctness
- c) Algorithm’s Design
- d) Both a & b

Question 54

“Loop Invariant” can be defined for

- a) Iterative Algorithms
- b) Recursive Algorithm
- c) Both a & b
- d) None of above

Question 55

“Loop Invariant” can be written in

- a) Descriptive Statement
- b) Mathematical Expression
- c) Both a & b
- d) None of above

Question 56

“At the start of each iteration of the for loop the subarray $A[1 \dots j-1]$ consist of the elements originally in $A[1 \dots j-1]$ but in sorted order.” is the loop invariant for:

- a) Merge Sort
- b) Insertion Sort
- c) Quick Sort
- d) None of above

Question 57

“Recurrence” is described as

- a) Equation / Equality
- b) Inequality
- c) Both a & b
- d) None of above

Question 58

“Recurrence” is used for describing the running time of:

- a) Iterative Algorithms
- b) Recursive Algorithm
- c) Both a & b
- d) None of above

Question 59

Which of the following is / are correct for “Recurrence” $T(n) = a T(n/b) + f(n)$

- a) $T(n/b)$ define the running time of a sub-problem
- b) “a” are the number of sub-problems
- c) $f(n)$ describe the running time of all statements in algorithm except recursive calss.
- d) All of above

Question 60

Which of the following is / are incorrect for “Substitution Method”

- a) Can solve upper and lower bounds simultaneously
- b) Powerful method to solve recurrences
- c) demands practice / expertise
- d) Use Mathematical Induction to prove the guess

Extra Sheet for Rough Work